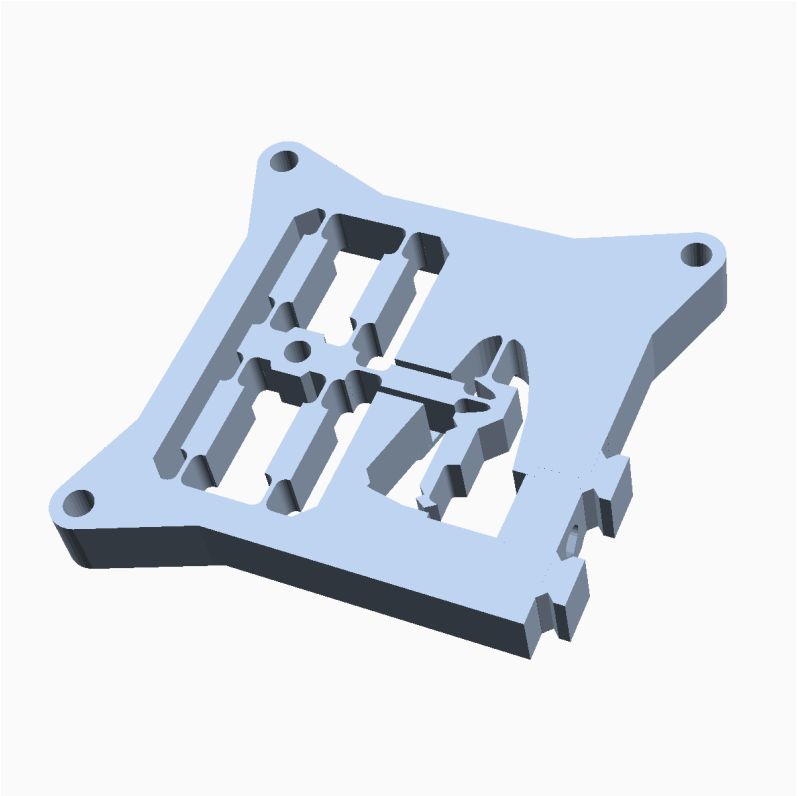
Abstract

This is the overview for the **compliant fine adjust**, which is an implementation for the **prismatic fine adjust** specification. It aims to:

- clarify the meaning of the design requirements
- inform the user on the design rules used to check the final design
- provide information on suggested 3D printing settings



WARNING: Do not use this component in safety critical application or any application which might cause harm to people and/or animals.

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Overview

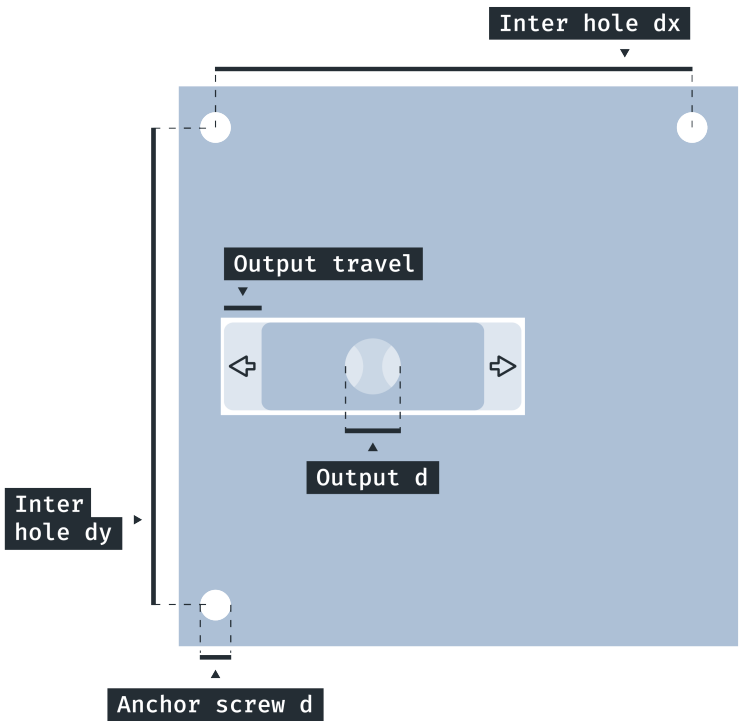
This compliant fine adjust mechanism allows you to precisely control the translation of the output shuttle from a rotational input. It can be viewed as a prismatic joint with no backlash actuated by a lever mechanism, which amplifies precision.

A few key points:

- Precise control with no backlash
- Print in place, no assembly required
- Easy to print (no draft nor support needed)

NB: This is still a work in progress. Your feedback/comments will be much appreciated! You can reach me at sol.info@gmail.com

Notes on key design requirement



- **Output travel** Required motion of output shuttle from the rest position. The complete range of motion will be double this value, going from *negative* **Output travel** to *positive* **Output travel**.
- **Gain** Precision amplification factor. The greater this value, the more precise the mechanism. A motion of the input lever of 1 mm will result in an output motion of 1/**gain** mm.
- **Output d** The diameter of the interface hole in the output shuttle.
- **Anchor screw d**: Size of the screws which will held the device in place. Note that a suitable tolerance will be added automatically, so the holes on the device will fit standard screws.
- **Thickness max** Maximum thickness which the devide can occupy.
- **Load max** Maximum vertical load which must be supported by the output shuttle.
- **Cycles** How many cycles the component should be able to survive without failures. This is specification is applied to all compliant hinges.

Design considerations

SOL75 chooses the values of a set of design parameters so that, when considered with the requirements, they satisfy a given set of constraints. Here we review the most interesting of these constraints. This should clarify what the component *has been designed for*, and how suitable it may be for your application.
Note that some constraints, like bounds on mass or temperature, will not be commented here, as they should be self explanatory.
If you find any mistakes, please let me know at sol.info@gmail.com

Overview of main design constraints:

- **Vertical load** The vertical load is applied to the output shuttle, which is supported by 4 identical links. These share the vertical load equally. The weakest part of each link is the hinge which connects it to the frame. The maximum stress is the hinge can be estimated as

$$\sigma_{\text{max}} = \frac{M}{I}\left(\frac{h}{2}\right) = \frac{{}^{F}/_{4} \cdot l_{\text{link}}}{I}\left(\frac{h}{2}\right)$$

  Where $I = h^3 \cdot {}^{t}/_{12}$ for the rectangular cross section, We impose that this stress should be less than 85% of the Ultimate Tensile Strength for the material (the value are reported in Table **??**).

- **Fatigue life of hinges** The design uses two different hinges, the ones for the links, which undergo a small deflection, and the one on the lever, which is subject to a larger one. We impose for the maximum deflection stress to be below the SN stress curve for the material.

$$\sigma(\theta) < \sigma_{\text{Material}}(k_{\text{cycles}}) \tag{1}$$

  The stress in a beam from the deflection can be approximated as

$$\sigma(\delta) = \frac{3E\delta}{l^2} \cdot \left(\frac{t}{2}\right)$$

  In this case, l is the lenght of the hinge, t its thickness, and E the Young's modulus of the material. We can express the deflection as $\delta = {}^{l}/_{2}\tan(\theta)$, where the hinge lenght is halved. This is because we assume that the equivalent hinge pivot point is in the middle of the compliant hinge.
  The design requirement is

$$\sigma(\delta) = \frac{3Et}{4l}\tan(\theta) \quad \leq \sigma_{\text{Material}}(k_{\text{cycles}}) \tag{2}$$

Mechanism kinematics

We can study the motion of the deamplifier with and equivalent rigid body model. The compliant hinges are considered like standar hinges, and everything else is assumed to be rigid. The resulting ridig body diagram is shown in Fig. 1.

First order equation: Moving the node **C** in by x_{in}, as shown in Fig. (1) turns the input lever **A-C** around the hinge point **A**. If the rotation θ is small, we can neglect the motion of the point **B** in the y direction. Then, **B** translates in x, pushing on the output shuttle (the gray rectangle in figure), which translates by the same amount.
Therfore, a first order *gain relationship* between input and output can be written as

$$x_{\text{output}} \approx \frac{l_0}{l_0 + l_1} \cdot x_{\text{input}} \tag{3}$$

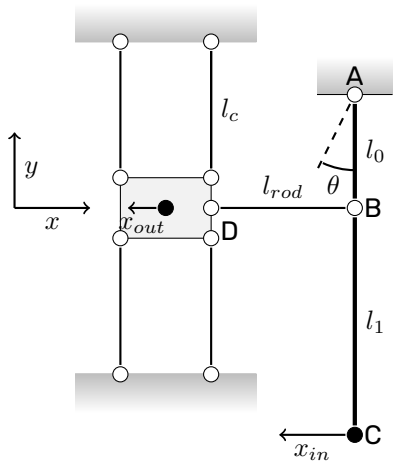


Figure 1: Rigid body model for the analysis of the deamplifier. Note that l_0 and l_1 are part of the same rigid body. Not to scale.

Full analysis: Assuming that the hinge behaves ideally, we can write the displacement of the point **B** as a function of the input angle θ as

$$\begin{cases} \Delta x_B &= l_0 \cdot \sin(\theta) \\ \Delta y_B &= l_0 \cdot (1 - \cos(\theta)) \end{cases}$$

Due to symmetry in the shuttle support structure, the point **C** can not move in the y. Assuming that the rod is rigid (its length stays the same), we write the position of point **D** as a function of point B

$$\begin{cases} x_C &= x_B - \sqrt{l_{rod}^2 - y_B^2} \\ y_C &= 0 \end{cases} \quad \rightarrow \quad \Delta x_C = (X_{B,0} - l_{rod}) - \left(X_{B,1} - \sqrt{l_{rod}^2 - y_{B,1}^2}\right)$$

Finally, using basic trigonometry, we can express the displacement of the output shuttle as a function of the input

$$\Delta x_{out} = \left(\frac{l_0}{l_0 + l_1}\right) \Delta x_{in} - l_{rod} \left[1 - \sqrt{1 - \left(\frac{l_0}{l_{rod}}\right)^2 \cdot (1 - \cos(\theta))^2}\right] \tag{4}$$

Therefore, the smaller the ratio l_0/l_{rod}, the better the linear approximation, reducing the drift effect caused by θ.

3D printing

Printing settings

Infill All stress equations assume 100% infill, so for heavy duty application, this would be the safest option. However, if it is just a test print or a non critical component which can easily be replaced, any infill can be used. Most geometries feature very thin element, so the actual infill percentage will not matter much.
This being said, stress and fatigue equations are taken directly from handbooks and scientific papers, which usually refer to solid materials. If you print with reduced infill and carry out fatigue tests, please let me know the results!

Orientation The model in the STL files are already in the suggested print orientation. The preview images, like the one in the first page of this document, depict the assembled configuration.

From ".scad" to ".STL"

If you have chosen to download the 3D files in ".scad" format, you can more easily modify them, but you'll need to convert them to ".STL" in order to print them. Once you have openScad installed, the process is very easy:

1. Open the *A0_my_component.scad* file in the *source* folder with OpenScad
2. **Press F5** to view the model preview; you can now inspect it in 3D
3. **Press F6** to generate the 3D geometry for export (this might take a while depending on how complex the geometry is)
4. **Press F7** to export the STL (or choose it from the menu *File > Export > STL*)

About Openscad

STL files can be generated from the *.scad* files located in the *3D_files* folder. Note that *scad* files are really just text files, so you can open them with any text editor. However, to inspect the 3D model and to generate the STL file, you need to use **Openscad**. Openscad is a lovely, free and open source CAD software available for linux, windows and mac. You can download it from the official website www.openscad.org.

Details on model dependencies

Depending on the complexity of your project, the *3D_files* folder might contain many files. The root file will always be *A0_my_component.scad* (where *my_component* would be the name of the specific project). Sub-components used by the root will have the prefix *A0 An*, where n will be an increasing number. The m-th sub-components of the n sub of the root will have the prefix *A0 An Am*, and so on.

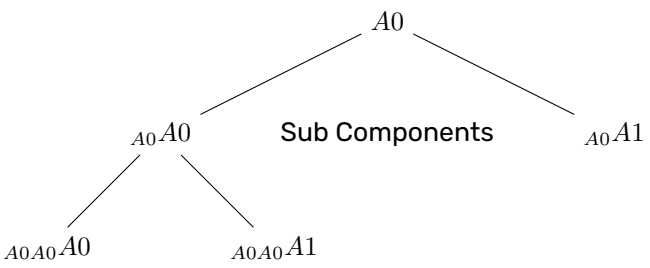


Figure 2: The prefix naming scheme reflects the component design tree. *A0* is always the root component.

Each *.scad* file will contain many modules[1]. However, components generated by SOL75 will always include four fundamental modules:

- *name_STAND_ALONE_PRINT()*: lays out all components in the suggested configuration for printing. Components which should not be printed (like a screw or an Arduino board) will not be included in this module.
- *name_NULL()*: displays the assembled geometry. You can use it to visually inspect clearances, mating surfaces and overall dimensions. **NB:** You should not print this configuration, as it might includes components which should not be printed or it might fuse together pieces which should have been printed separately.
- *name_POSITIVE()*: the component geometry which should be added to the parent geometry. The root might combine sub POSITIVE geometries to print them as a single piece.
- *name_NEGATIVE()*: this module presents the negative space which should be left for the correct operation of the component. Think of a spinning propeller; you need to have a large free cylindrical volume for it to work properly. When placing a component in your project, you should first remove this space to avoid collisions.

[1] In openScad, modules are special functions used to produce 3D geometries; they are defined by the keyword *module* followed by their name and parameters

Machine settings

The component has been designed for a machine with the performances detailed in Table **??**.

Parameter	Value	Definition
min w	0.5 mm	Minimum printable width. Minimum *thickness* which ensure a good quality print. This is the thinnest wall you can print.
nozzle d	0.4 mm	Nozzle diameter. If possible, features thickness will be chosen as a multiple of this value.
min layer t	0.2 mm	Minimum layer height. Minimum height which ensures good quality. Features shorter than this will not appear in the print.
min tol	0.4 mm	Minimum tolerance. The minimum value x such that a cylinder of diameter d is guaranteed to fit in a hole of diameter $d + x$.
pip tol	0.6 mm	Print-in-place tolerance. The minimum value x such that a cylinder of diameter d, printed in place within a hole of diameter $d + x$, will rotate freely.
max bridge	30 mm	Max allowable bridge length.
max overhang	45 °	Max allowable overhang angle (from the vertical).
printer bed size x	180 mm	One size of the printing bed.
printer bed size y	180 mm	The other size of the printing bed.
printer max h	180 mm	Maximum height for a printed part.

Table 1: 3D printer settings used in the design.

Material properties

The material chosen for this component is Generic PETG plastic. Table **??** reports its assumed properties, which have been used in the design.

Parameter	Value	Explanation
ρ	1380 kg/m^3	density
E	2000 MPa	Young's modulus
YS	50 MPa	Yield Strength
UTS	55 MPa	Ultimate Tensile Strength
min temp	-20 °C	Minimum safe operating temperature
max temp	80 °C	Maximum safe operating temperature

Table 2: Main assumptions about the material.

Legal Notice

Plain English

All models are wrong, but some are useful

- SOL75 is not a substitute for proper testing and prototyping
- Do not use any component generated by SOL75 for safety critical applications

Although I try to use the best available models, I can not guarantee the implementation to be error free. Moreover, uncertainties regarding the material properties, the 3D printing process and the application are just too big to be able to ensure any safety.

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